

Advanced Natural Language Processing

Applications of ANLP Abstracting, Machine Translation & Chatbot

Department of Advanced Computer Sciences (DACS)
Master of AI / DSDM
Prof dr ir Jan C. Scholtes – Course year 2026-2027



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Abstracting - Summarization



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Techniques to create Summaries

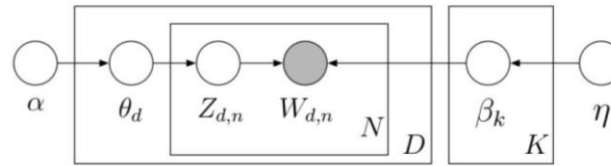
- **Extractive summarization** merely copies informative fragments from the input.
 - Sentence selection
 - Sentence picking keywords, key phrases, unsupervised (Latent Dirichlet Allocation – LDA or Non-Negative Matrix Factorization - NMF), supervised (Conditional Random Fields - CRF)
 - Variables: length (% or absolute # sentences), key phrases
- **Abstractive summarization** may generate novel words.
 - Generative Grammars – this never really worked well...
 - Deep Learning Decoders:
 - Very successful
 - Approach summarization as a “sequence-to-sequence” task: input is the original sequence, abstract is the output sequence

Extractive Summarization Approach

- Find small windows in doc that contain query terms
 - Requires fast window lookup in a document cache
- Score each window with query
 - Use various features such as window width, position in document, etc.
 - The *best* abstract sentences are determined by calculating various feature values for each sentence and then take the most probable sentences.
 - Combine features through a scoring function
- These algorithms can be tuned by setting the maximum length of a summary as percentage of the text or as a maximum number of words.
- Various open-source algorithms are available. (We will try one in the tutorial).

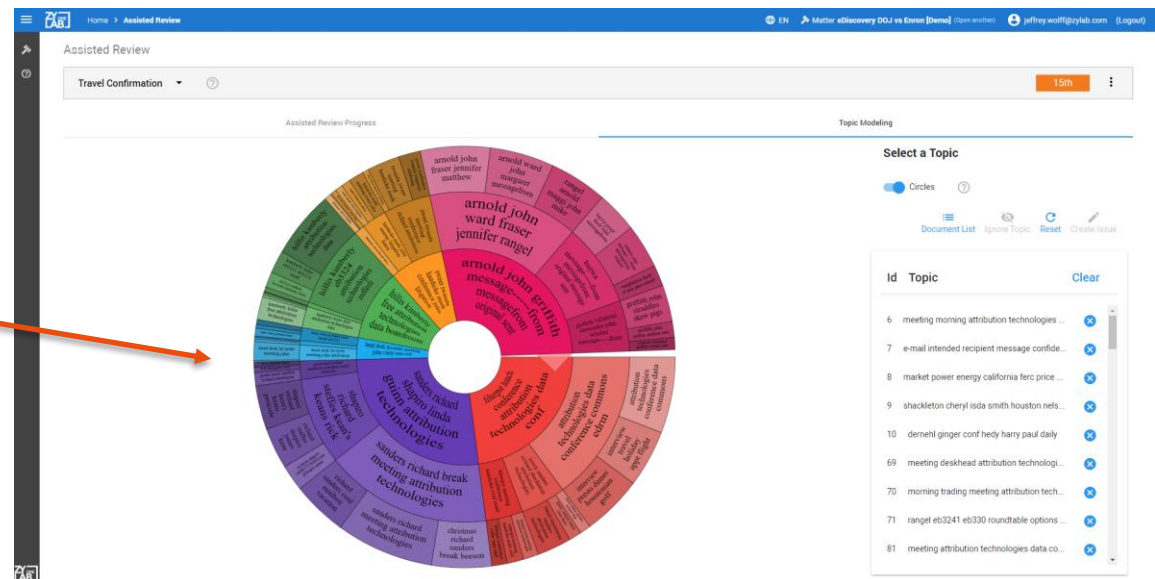
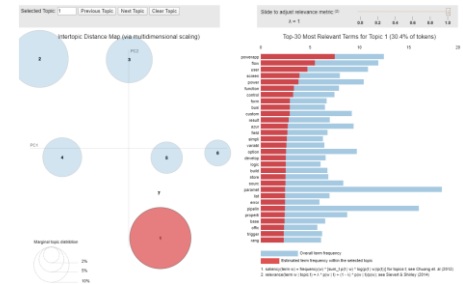
Key Phrases

- Key phrases can be identified by using search and clustering (topic modeling) approaches:
- Key words (search)
 - Key phrases (search)
 - Topic Modeling (discussed in Information Retrieval and Text Mining course) such as LDA and NMF
 - Supervised using Conditional Random Fields (CRF).



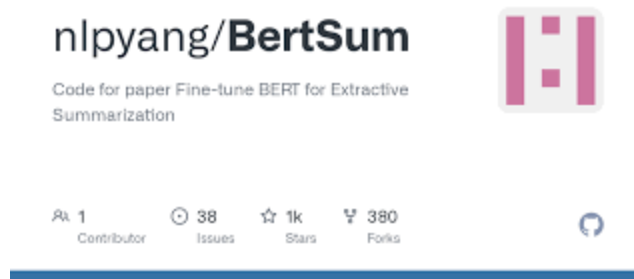
K – total number of topics
 β_k – topic, a distribution over the vocabulary
 D – total number of documents
 θ_d – per-document topic proportions
 N – total number of words in a document (it fact, it should be N_d)
 $Z_{d,n}$ – per word topic assignment
 $W_{d,n}$ – observed word
 α, η – Dirichlet parameters

- Several **inference algorithms** are available (e.g. sampling based)
- A few **extensions** to LDA were created:
 - Bigram Topic Model



Abstracting with BERT (or actually BERTSUM)

- Does this make sense? BERT is only the encoder model of the Transformer. How can we use this for generating abstracts?
- Abstracts come in two categories: extractive and generative.
- BERTSUM (the fine tuned BERT model for summarization) can be used best for extractive, but with some tweaking, it can also be used as part of a generative abstractor.



BERTSUM: Differences compared to BERT

- The BERTSUM model which is an extension of the BERT model but in particular to the task of text summarization. The main difference between BERT and BERTSUM is the addition of inputting data with symbols to represent start and end of sentence so that the model may learn sentence representations.
- Additionally, another difference is in the segment embeddings and how BERTSUM embeds pairs of sentences to learn adjacency patterns between each input sentence.
- The overall goal of this model is to give every sentence in the document a score representing the relevance of the sentence to the overall document, as a way to indicate to the model which sentence should be included in the summary.



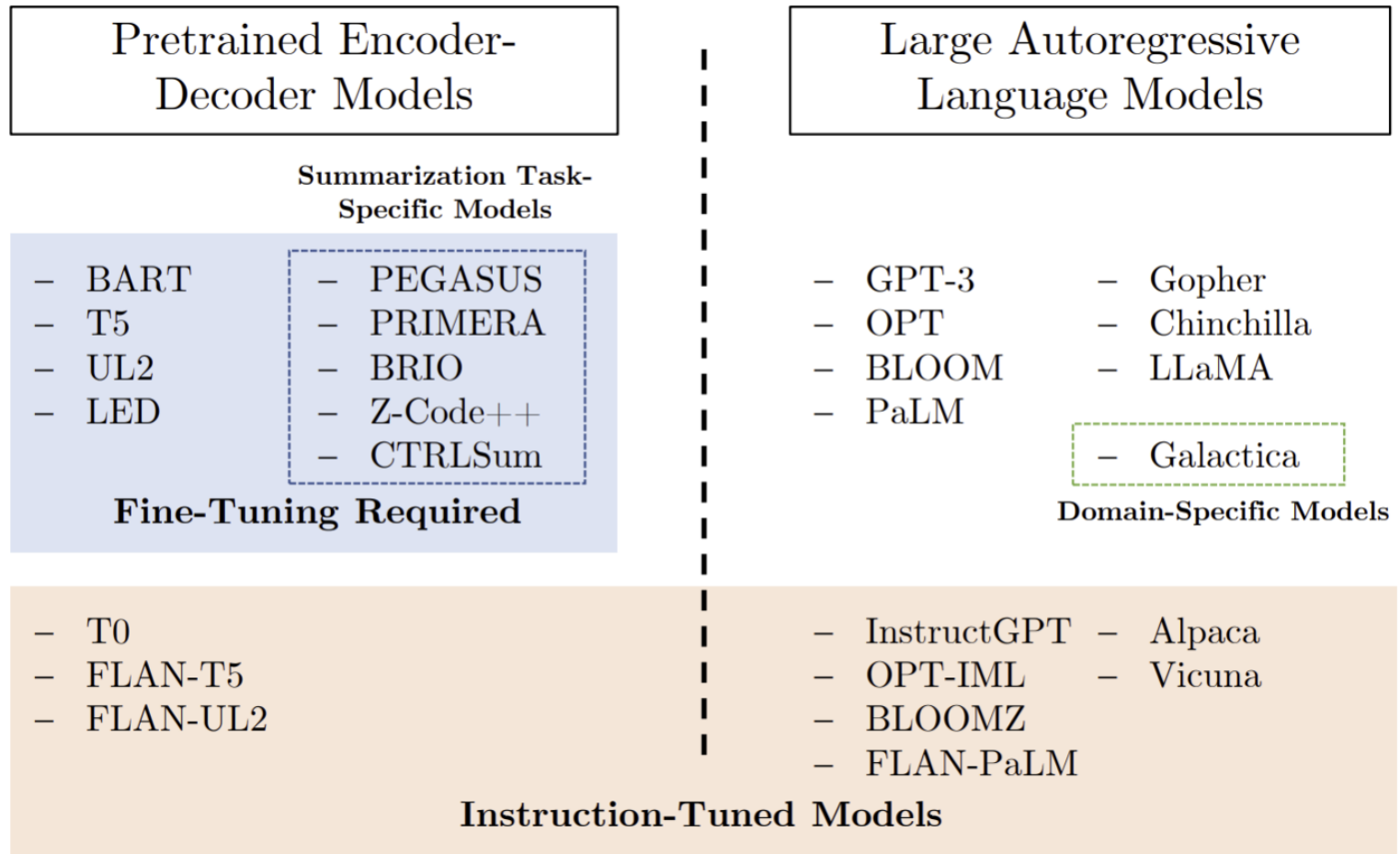
EXTRACTIVE!

Generative Summarization: using text generation

- Approach summarization as a “sequence-to-sequence” task: input is the original sequence, abstract is the output sequence
- Challenges:
 - Decoding is done iteratively and requires significantly more steps than the encoding
 - Quality and diversity of generated text depends on both decoding method and associated deep-learning (hyper) parameters:
 - Greedy search
 - Beam Search
 - Sampling: random, top-k, nucleus



Abstracting and Transformers



Abstracting with T5

- Use the encoder for the original text
- Use the decoder for the abstract

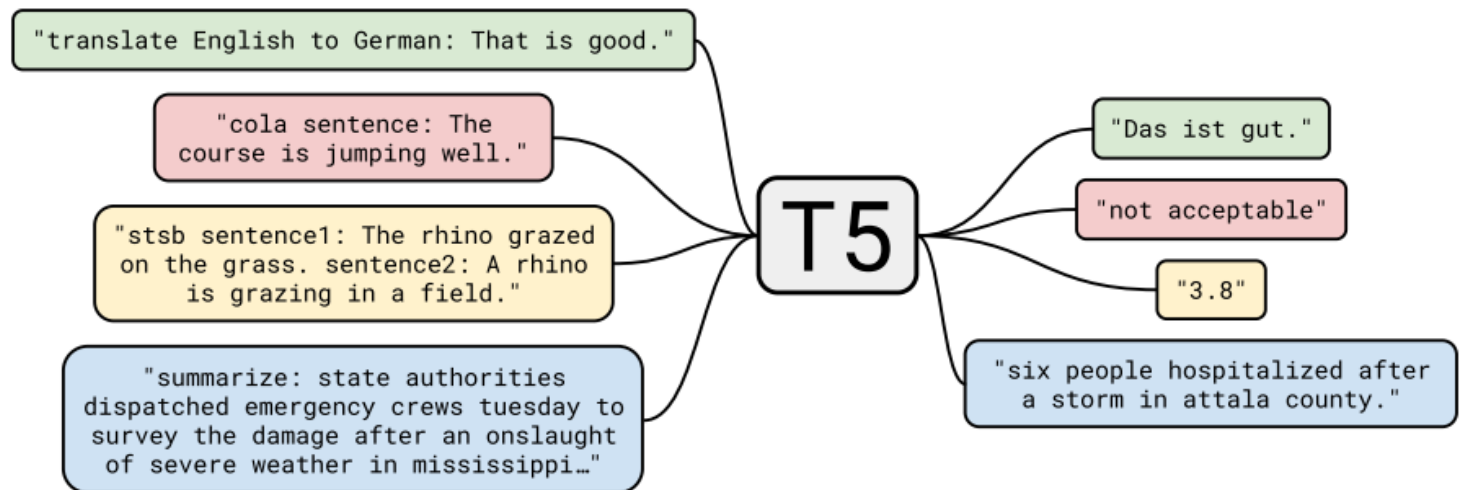
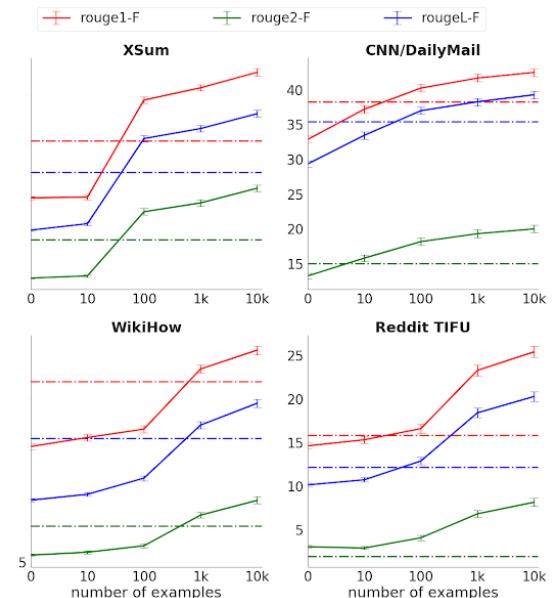
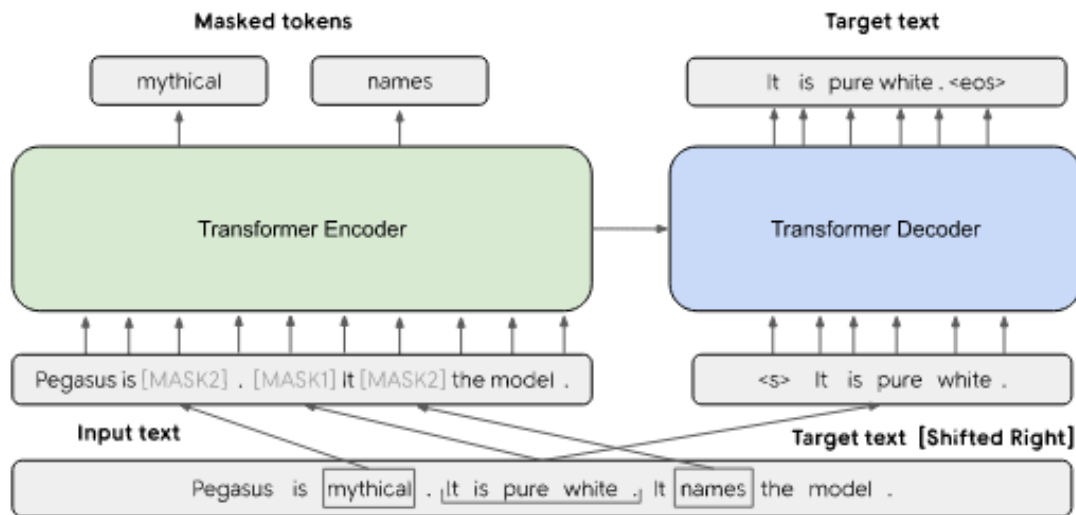


Figure 1: A diagram of our text-to-text framework. Every task we consider—including translation, question answering, and classification—is cast as feeding our model text as input and training it to generate some target text. This allows us to use the same model, loss function, hyperparameters, etc. across our diverse set of tasks. It also provides a standard testbed for the methods included in our empirical survey. “T5” refers to our model, which we dub the “**T**ext-**t**o-**T**ext **T**ransfer **T**ransformer”.

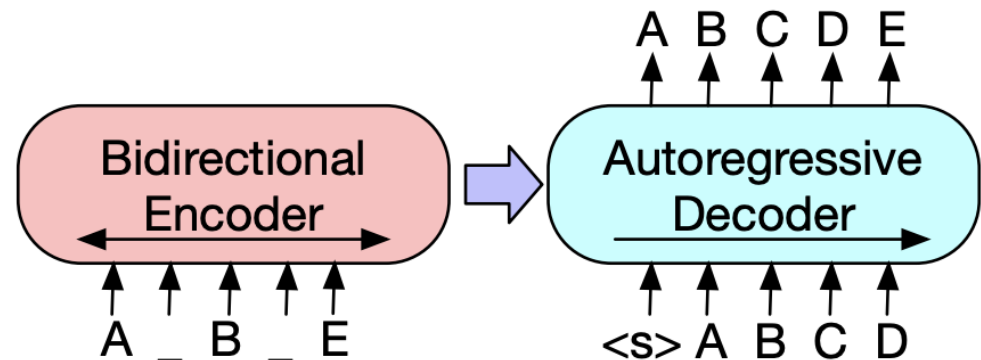
Abstracting PEGASUS

- PEGASUS: Google's State of the Art Abstractive Summarization Model.
- In "PEGASUS: Pre-training with Extracted Gap-sentences for Abstractive Summarization": uses a pre-training self-supervised objective (called gap-sentence generation) for Transformer encoder-decoder models to improve fine-tuning performance on abstractive summarization, achieved state-of-the-art results on 12 diverse summarization datasets.



Abstracting with BART

- BART is a denoising autoencoder for pretraining sequence-to-sequence models. It is trained by
 - (1) corrupting text with an arbitrary noising function, and
 - (2) learning a model to reconstruct the original text.
- It uses a standard Transformer-based encoder-decoder translation architecture with a bi-directional encoder (like BERT) and a left-to-right decoder (like GPT).
- This means the encoder's attention mask is fully visible, like BERT, and the decoder's attention mask is causal, like GPT2.
- Original text in encoder, abstract from decoder.



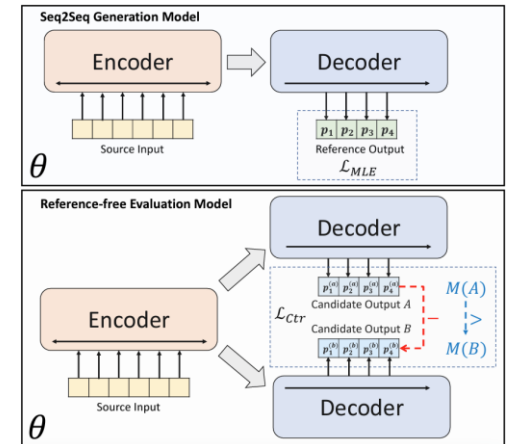
System	Summary
Reference	chelsea forward tammy abraham nets first-half double for chelsea. dominic solanke adds a third late on as chelsea look set to win trophy. manchester city struggle without injured star thierry ambrose. read: mourinho warns his young chelsea players he can not play them all. click here to read our match report from man city 's academy stadium.
BART	tammy abraham scored twice in the first half to give chelsea the lead. isaac buckley-ricketts levelled the game for manchester city. dominic solanke scored late on to put a gloss on the scoreline. click here to read sportmail's player ratings from the youth cup final.
BRIO-Mul	chelsea beat manchester city 3-1 in the youth cup final at the etihad stadium. tammy abraham scored twice in the first half to give chelsea the lead. dominic solanke scored late on to seal the win for the home side.
Reference	alejandro valverde won ahead of julian alaphilippe and michael albasini. chris froome finished 123rd after a crash during the final 12 kilometres. team sky's sports director gabriel rasch praised froome for finishing. rasch said froome was 'banged up' but expects to ride tour de romandie.
BART	movistar rider alejandro valverde won fleche wallonne on wednesday. team sky's chris froome fell in the final 12km but finished the race. philippe gilbert pulled out of the race after a bad crash 50km from the end. click here for more cycling news.
BRIO-Mul	alejandro valverde defended his fleche wallonne title in belgium on wednesday. movistar rider finished ahead of julian alaphilippe and michael albasini. team sky's chris froome fell in the final 12km of the race but finished in 123rd. froome was involved in a crash but finished the race despite being 'banged up'
Reference	manuel pellegrini won the premier league and capital one cup last season. city currently sit fourth in the league table - 12 points behind chelsea. pellegrini's contract expires at the end of the 2015-16 season. city players have been impressed with vieira's work with the youth team. pep guardiola is city's first-choice to succeed pellegrini at the etihad.
BART	manuel pellegrini's future at manchester city is under scrutiny. patrick vieira is highly-respected among the city players. city's first-choice managerial option is bayern munich boss pep guardiola. click here for all the latest manchester city news. click here for more premier league news.
BRIO-Mul	manchester city players have backed patrick vieira to replace manuel pellegrini as manager of the club. the frenchman is highly-respected among the players at the etihad stadium. pellegrini's future at the club is under scrutiny after a disappointing season. city's first-choice manager is current bayern munich boss pep guardiola.

Table 10: Case Study on CNNDM. BRIO-Mul learns to ignore the noise pattern ("click here") while BART cannot.

Yixin Liu, Pengfei Liu, Dragomir Radev, and Graham Neubig. 2022. BRIO: Bringing Order to Abstractive Summarization. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 2890–2903, Dublin, Ireland. Association for Computational Linguistics.

System	R-1	R-2	R-L	Acc.(%)
High	53.99	29.85	51.12	100.00
Low	33.48	10.85	30.45	0.00
BART	44.88	21.68	41.92	54.80
Ours	50.10	26.29	47.19	79.63

Table 1: Accuracy of different abstractive summarization systems w.r.t ranking the quality of candidate summaries on CNNDM dataset. **Acc.** stands for the frequency of the model assigning higher probabilities to better candidate summaries. The candidate summaries are generated by a pre-trained model (BART), and we select the best and the worst candidates (w.r.t. ROUGE scores) for each of the samples. **High** and **Low** represent the average performance of the best and worst candidates respectively. R-1/2/L are the ROUGE-1/2/L scores. The original BART only achieves 54.80% accuracy.



Summarization with GPT

CNN article: <https://www.cnn.com/2022/09/09/politics/judge-throws-out-trumps-rico-lawsuit-against-hillary-clinton-and-democrats/index.html>

Length-constrained prompt

Summarize the above article in 2 sentences.

GPT-3: Donald Trump's lawsuit against Hillary Clinton, the Democratic National Committee, and several other individuals and entities has been dismissed by a federal judge. The judge ruled that Trump's claims are unsupported by any legal authority and that many of his specific factual assertions are implausible or unsupported.

Keyword-constrained prompt

Summarize the above article briefly focusing on Alina Habba.

GPT-3: Alina Habba is an attorney for Donald Trump. She stated that Trump will "immediately move to appeal this decision." Habba also said that the judge's opinion is "rife with erroneous applications of the law" and that it "disregards the numerous independent governmental investigations which substantiate our claim that the defendants conspired to falsely implicate our client and undermine the 2016 Presidential election."

Figure 3: Exemplary instructions for zero-shot summarization using GPT-3. Notably, the natural language instructions of LLMs enable greater control over tasks, such as length-constrained summarization. (Goyal et al., 2022)

Sticking to the facts: Summarization of Medical Documents

“Our findings reveal that LLMs could be susceptible to generating factually inconsistent summaries and making overly convincing or uncertain statements, leading to potential harm due to misinformation.

Moreover, we find that models struggle to identify the salient information and are more error-prone when summarizing over longer textual contexts”

Tang, L., Sun, Z., Ilday, B. *et al.* Evaluating large language models on medical evidence summarization. *npj Digit. Med.* 6, 158 (2023).
<https://doi.org/10.1038/s41746-023-00896-7>

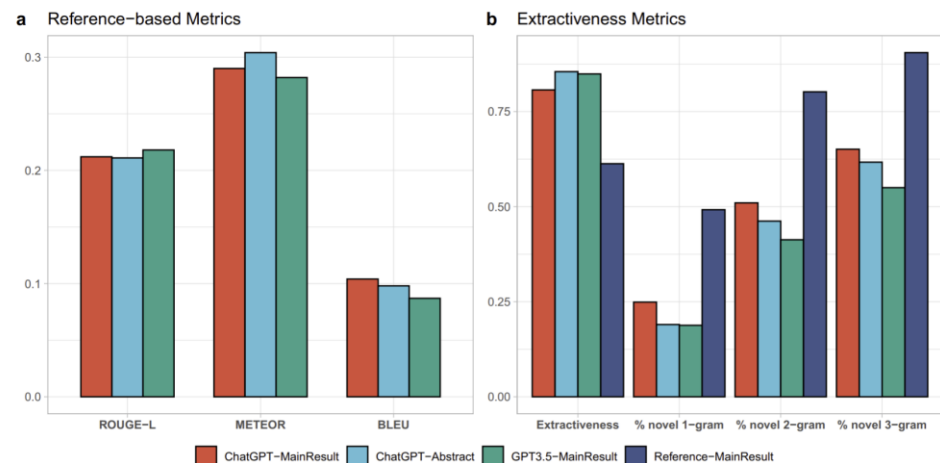
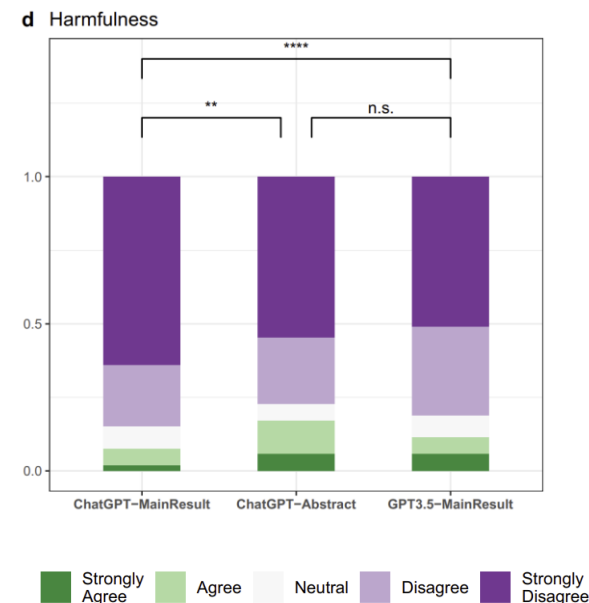


Fig. 1 Performance of different summarization systems in automatic evaluations. **a** Reference-based metrics (higher scores indicate better summaries). **b** Extractiveness metrics.



Machine Translation

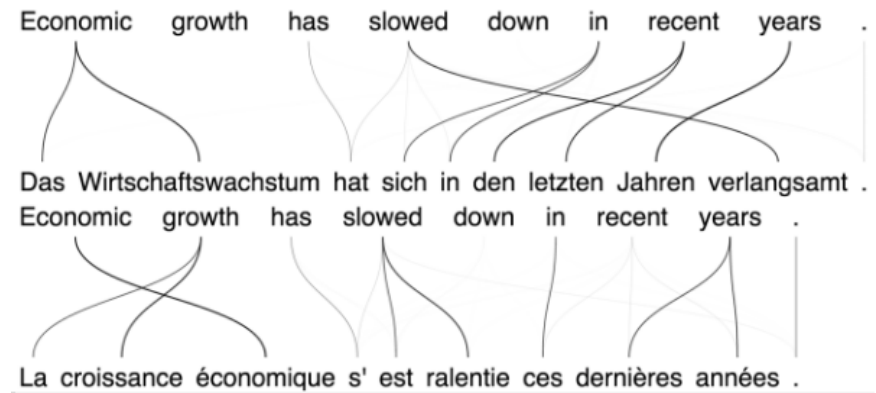


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Machine Translation Challenges

Why is machine translation difficult?

- Alignment (word order) →
- Lexical ambiguity →
- Syntactic ambiguity
- Semantic ambiguity (synonym, hyponym, homonym)
- Idiom & Proverbs
- Co-references
- Pronouns
- Negations
- New words (Brexit)
- Slang
- Language mixtures (“downloaded”)
- Errors



- English: fish → Spanish: pez, pezcado
- English: rice → Japanese: cooked-rice, plant, ...

Early Approaches in Machine Translation

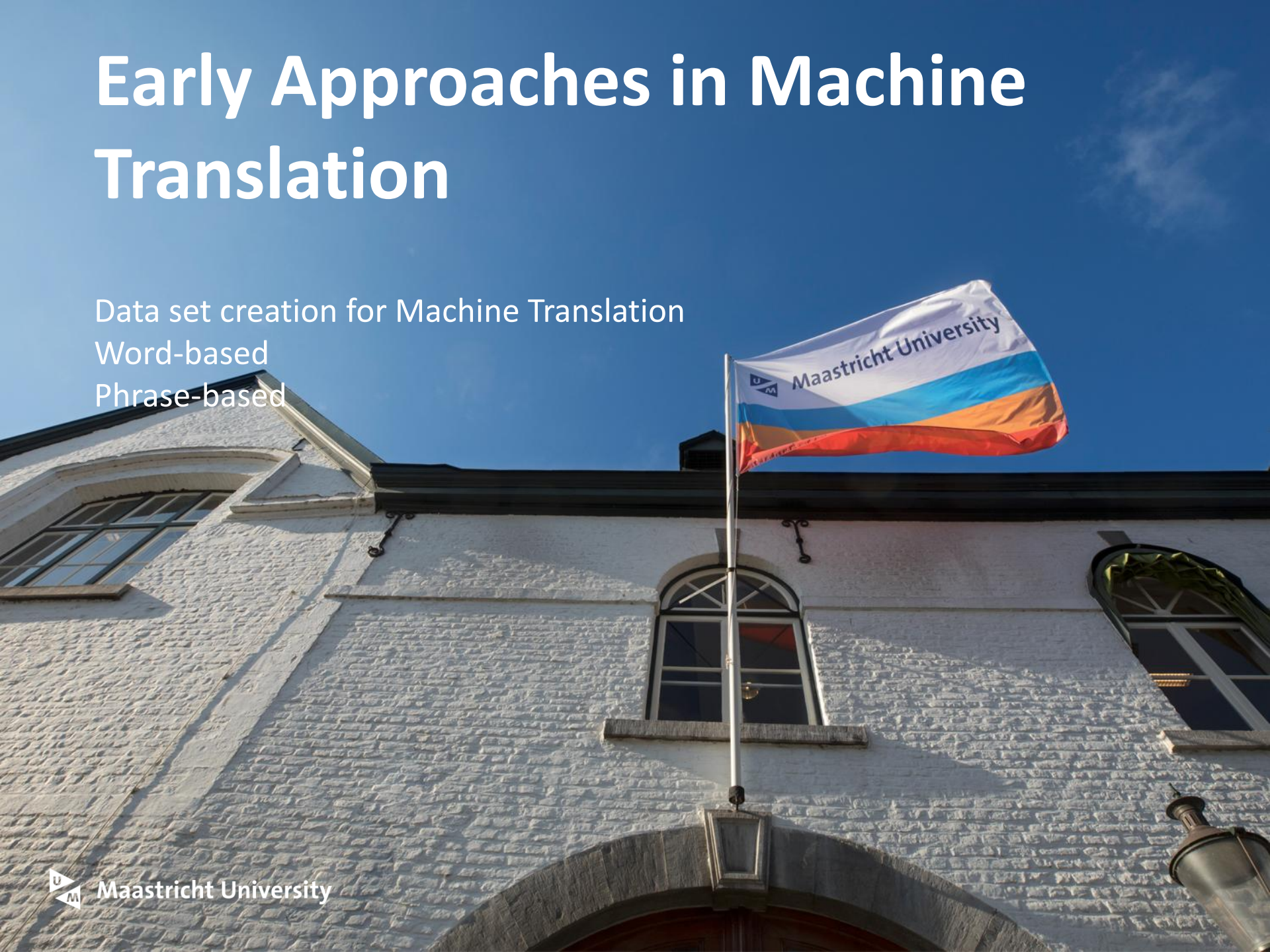
Data set creation for Machine Translation

Word-based

Phrase-based

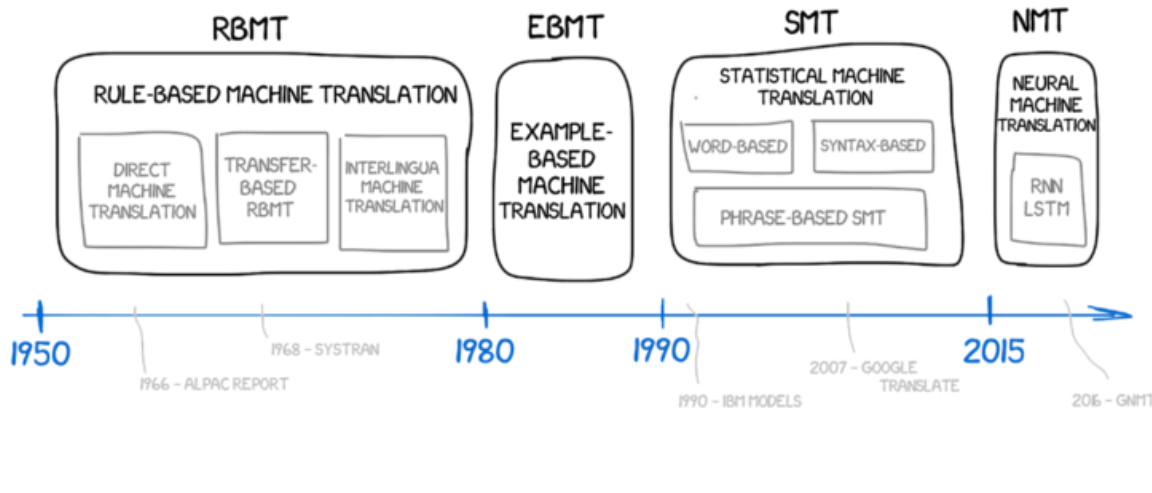


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Machine Translation: brief history

A BRIEF HISTORY OF MACHINE TRANSLATION



THE ORGANIZATION OF THE ROSETTA GRAMMARS

Jan Odijk

Philips Research Laboratories, P.O. Box 80.000
5600 JA Eindhoven, The Netherlands

ABSTRACT

In this paper the organization of the grammars in the Rosetta machine translation system is described and it is shown how this organization makes it possible to translate between words of different syntactic categories in a systematic way. It is also shown how the organization chosen makes it possible to translate 'small clauses' into full clauses and vice versa. The central concept worked out here in some detail is the concept of 'partial isomorphism' between subgrammars. The system as described here has been implemented and is currently being tested.

1 ROSETTA

In this section I will mention some essential properties of the Rosetta machine translation system. For more extensive descriptions of this system I refer to Landberger(1987), Appelo & Landberger(1988), Appelo, Föllinger & Landberger(1987) and Leemakers & Roux(1986).

The Rosetta machine translation system is an interlingual machine translation system based on Montague Grammar and developed at Philips Research Laboratories in Eindhoven for Dutch, English and Spanish. The grammars used in Rosetta are computational variants of Montague Grammar and are called M-grammars.

Rules in M-grammars, called M-rules, operate on syntactic surface trees. If a certain string *str* is associated with a syntactic tree of category *C* the notation *C(str)* will be used.

An M-grammar in Rosetta is subdivided into a number of subgrammars. Each subgrammar is a rule package and is defined by specifying a *head* (the category of a designated argument from the import for the subgrammar), *export* (the categories of the syntactic trees yielded by the subgrammar)

and a *control expression* that indicates in which order M-rules must be applied.

In Rosetta³ a distinction is made between meaningful M-rules, called *rules* (that correspond to a meaning operation), and meaningless rules called *transformations* (that do not correspond to a meaning operation).

M-grammars are reversible. Because of this fact the translation relation between two sentences can be illustrated by showing their derivations in generation. The example derivations below will be restricted to generation only.

Within the Rosetta system the grammars of the relevant languages are attuned to one another in such a way that two sentences that are each other's translation can be derived starting from corresponding basic expressions by applying corresponding rules in the same way for both sentences 'Corresponding' here means: 'corresponding to the same meaning operation or basic meaning', and it holds of meaningful M-rules only.

M-grammars attuned in this way are called *isomorphic grammars*, and therefore the method of translation used is called the *isomorphic grammar approach to machine translation*.

2 CATEGORICAL MISMATCHES

In order to derive simple sentences e.g. *Ziet hij de man?* one could design a grammar of Dutch that starting with a basic verb (*zien*) applies rules to derive the sentence mentioned. One could have rule combining a basic verb with arguments that it allows or requires (*hij, de man*) in a sentence, rule that determines the tense of this sentence (*presens tense*) and rules that determine 'mood' and real

³The paper deals with the grammars in the third version of the Rosetta system, Rosetta³.

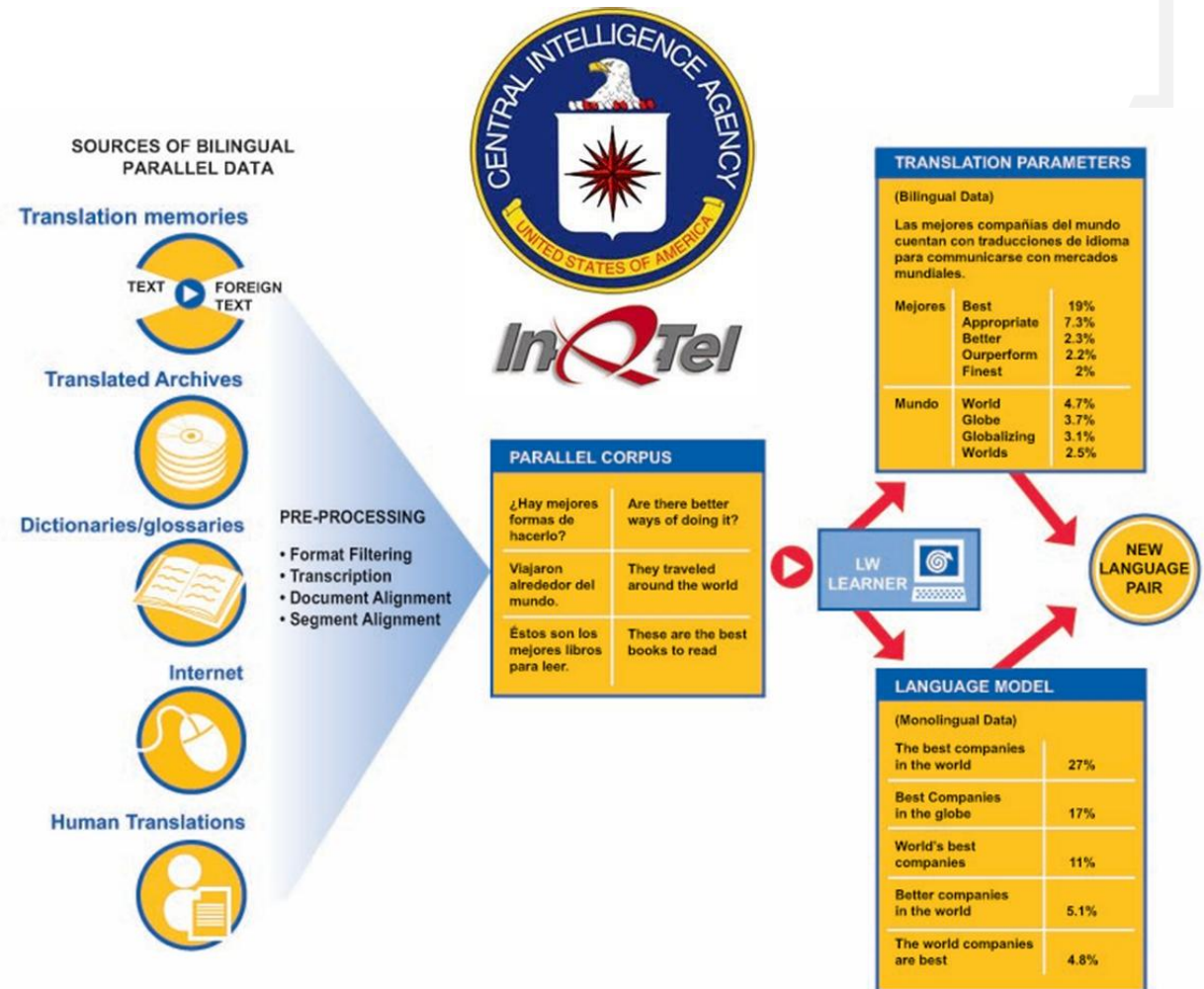
Source: <https://www.freecodecamp.org/news/a-history-of-machine-translation-from-the-cold-war-to-deep-learning-f1d335ce8b5/>

Source: <https://aclanthology.org/www.mt-archive.info/70/IAI-1986-Appelo.pdf>



Machine Translation: Intelligence Agencies

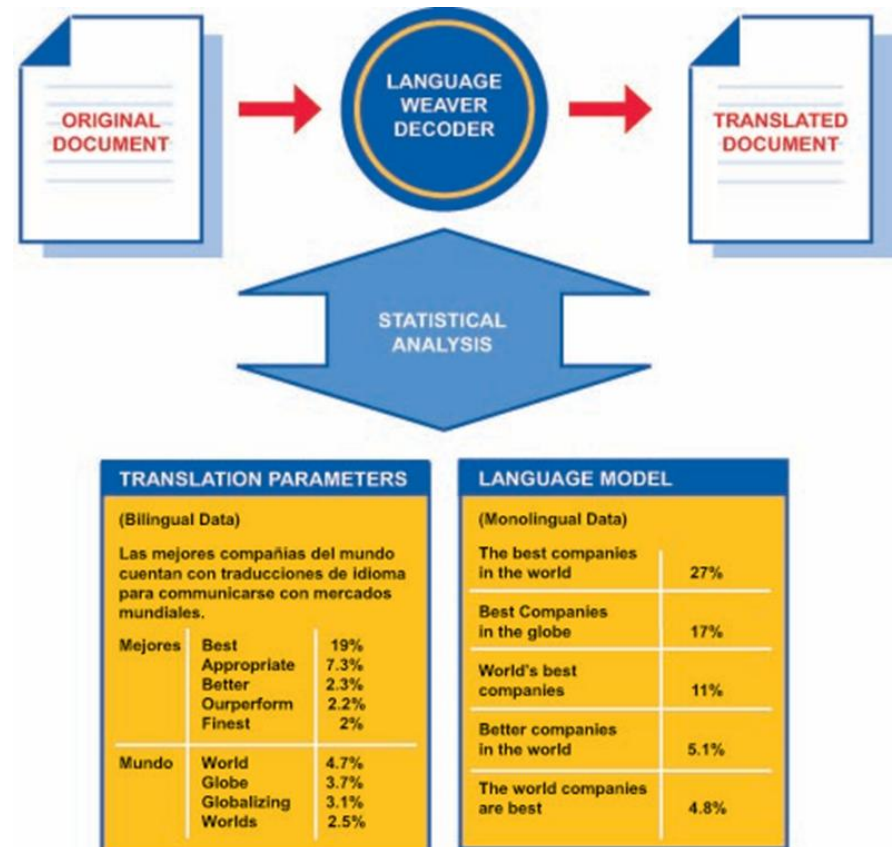
- First there was the Linguistic Approach. This never reached acceptable results
- Then came Language Weaver with a Corpus (aka Example) Based (statistical approach) – Now SDL



See also: https://en.wikipedia.org/wiki/Language_Weaver and <https://en.wikipedia.org/wiki/In-Q-Tel>

Statistical (Corpus-Based) machine Translation

- Highly robust
- Trainable by end users without deep understanding of the inner workings of the system
- Human-level quality within specific (fine-tuned) domains



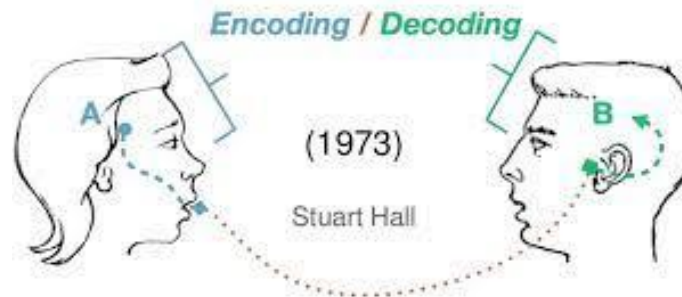
Machine Translation & Deep Learning: RNN and LSTM



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Encoding - Decoding

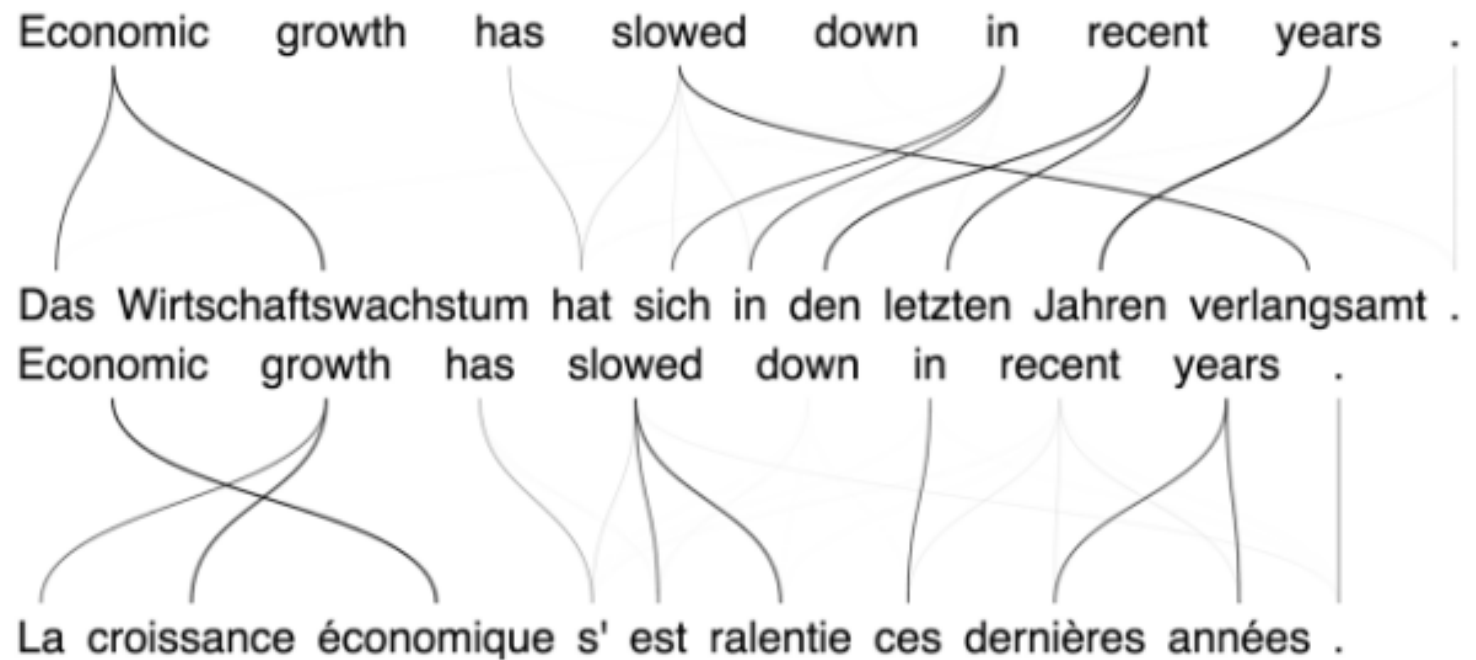
- Understanding language is an iterative process for humans: When we read or hear a sentence, we process each word accumulating information up to the end of the text. This can be seen as an encoding process.



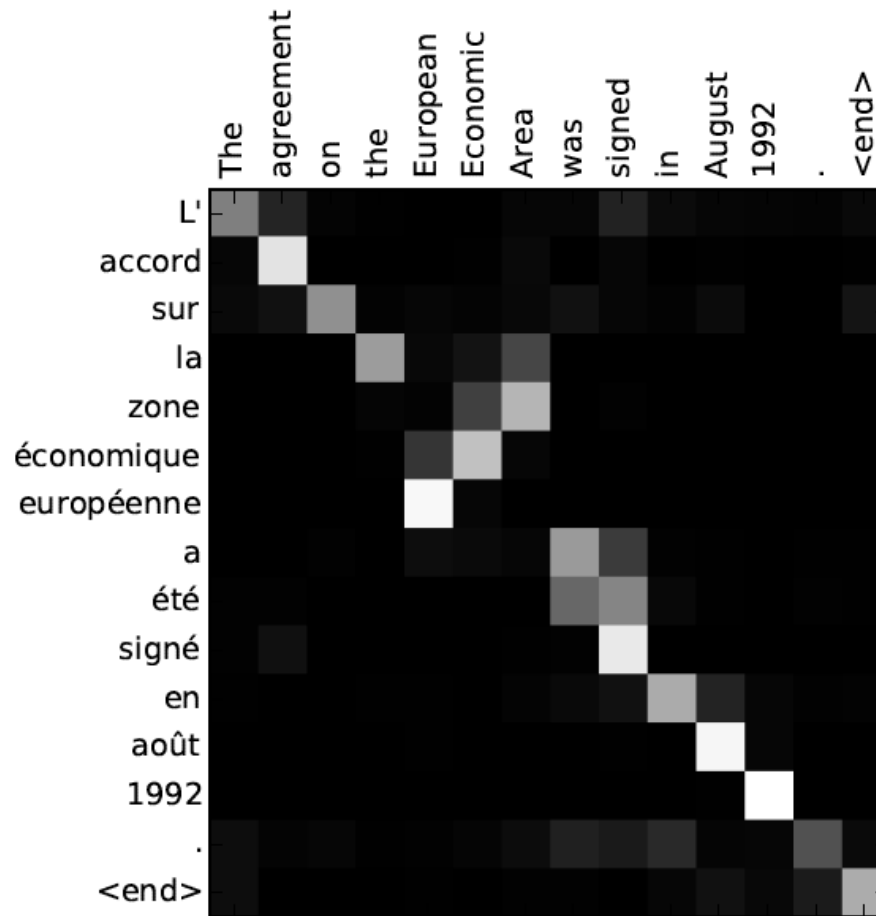
- Then, we perform a certain task: translation, Q&A, abstracting, story telling, ... These can all be seen as decoding processes.

Attention

- Idea:
 - Focus on different parts of the source sentence
 - Different parts are important for different target words
 - Alignment in SMT



BiRNN Encoder-Decoder with Attention



- Light is HIGH attention
- Dark is lower attention
- Black is no attention
- See how attention solves the alignment problem!

Transformers for Machine Translation: T5



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Transformers for Translation

- In the original Transformer paper “Attention is all you need” from Vaswani in 2017, the authors use a transformer-based encoder-decoder model for translation.

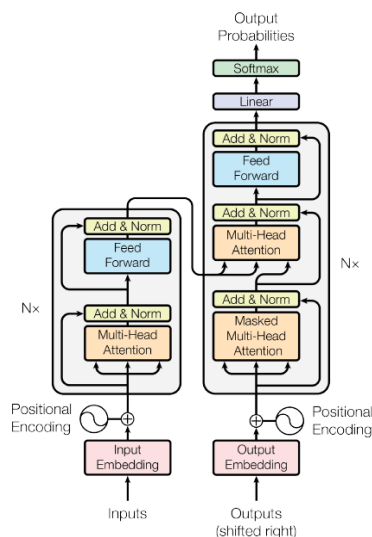


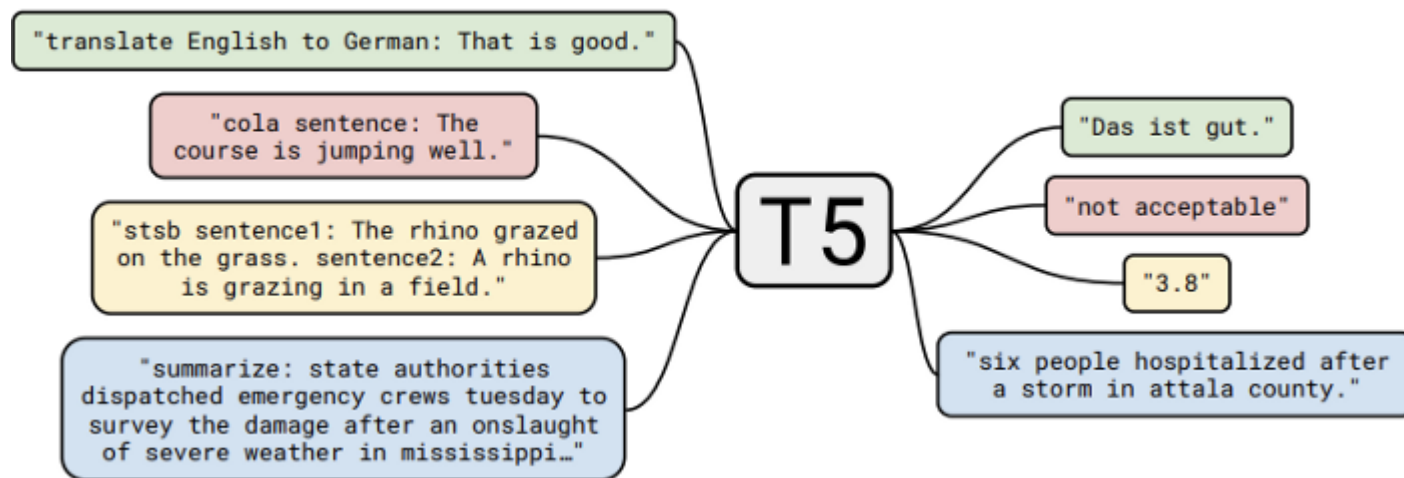
Figure 1: The Transformer - model architecture.

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

T5 for translation: a closer look

- T5 = Text-to-Text Transfer Transformer (T5)
- Encoder – Decoder model
- Transformer-based



T5

From the paper:

Transfer learning, where a model is first pre-trained on a data-rich task before being fine-tuned on a downstream task, has emerged as a powerful technique in natural language processing (NLP).

The effectiveness of transfer learning has given rise to a diversity of approaches, methodology, and practice. In this paper, we explore the landscape of transfer learning techniques for NLP by introducing a unified framework that converts every language problem into a text-to-text format.

Our systematic study compares pretraining objectives, architectures, unlabeled datasets, transfer approaches, and other factors on dozens of language understanding tasks.

By combining the insights from our exploration with scale and our new “Colossal Clean Crawled Corpus”, we achieve state-of-the-art results on many benchmarks covering summarization, question answering, text classification, and more. To facilitate future work on transfer learning for NLP, we release our dataset, pre-trained models, and code.

T5

- T5 is an encoder-decoder model pre-trained on a multi-task mixture of unsupervised and supervised tasks and for which each task is converted into a text-to-text format. T5 works well on a variety of tasks out-of-the-box by prepending a different prefix to the input corresponding to each task, e.g., for translation: *translate English to German: ...*, for summarization: *summarize:*
- T5 uses relative scalar embeddings. Encoder input padding can be done on the left and on the right.
- See the [training](#), [inference](#) and [scripts](#) sections below for all details regarding usage.
- T5 comes in different sizes: [t5-small](#) [t5-base](#) [t5-large](#) [t5-3b](#) [t5-11b](#).
- Based on the original T5 model, Google has released some follow-up works:
 - **T5v1.1:** T5v1.1 is an improved version of T5 with some architectural tweaks, and is pre-trained on C4 only without mixing in the supervised tasks. Refer to the documentation of T5v1.1 which can be found [here](#).
 - **mT5:** mT5 is a multilingual T5 model. It is pre-trained on the mC4 corpus, which includes 101 languages. Refer to the documentation of mT5 which can be found [here](#).
 - **byT5:** byT5 is a T5 model pre-trained on byte sequences rather than SentencePiece subword token sequences. Refer to the documentation of byT5 which can be found [here](#).

Pegasus

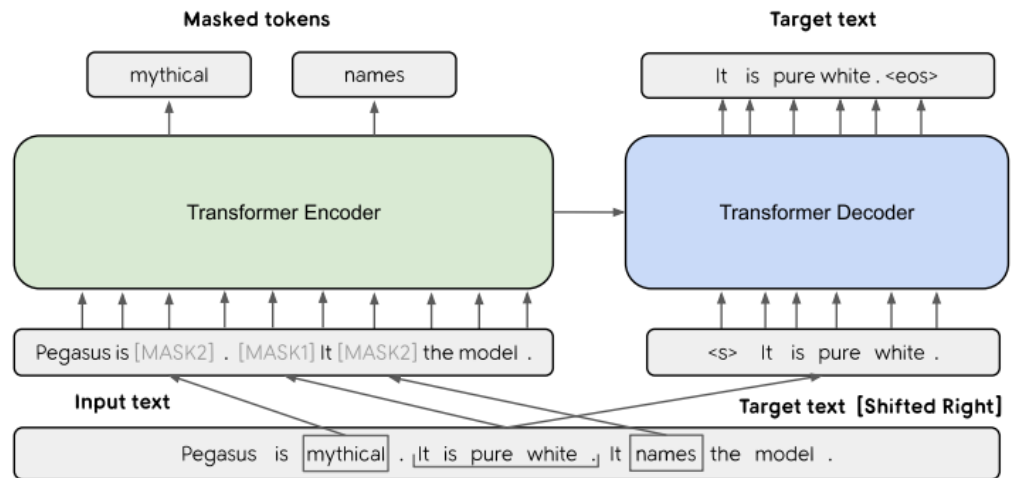


Figure 1: The base architecture of PEGASUS is a standard Transformer encoder-decoder.

- The Pegasus model was proposed in PEGASUS: “ Pre-training with Extracted Gap-sentences for Abstractive Summarization” by Jingqing Zhang, Yao Zhao, Mohammad Saleh and Peter J. Liu on Dec 18, 2019.
<https://arxiv.org/pdf/1912.08777.pdf>
- According to the abstract, Pegasus’ pretraining task is intentionally similar to summarization: important sentences are removed/masked from an input document and are generated together as one output sequence from the remaining sentences, similar to an extractive summary.
- Pegasus achieves SOTA summarization performance on all 12 downstream tasks, as measured by ROUGE and human eval.

Can BERT be used for Machine Translation?

BERT pretraining has two advantages over NMT training: it solves a simpler (monolingual) task of 'source sentence encoding', compared to NMT (bilingual task) which has to 'encode source sentence information', and 'translate into a target language'.

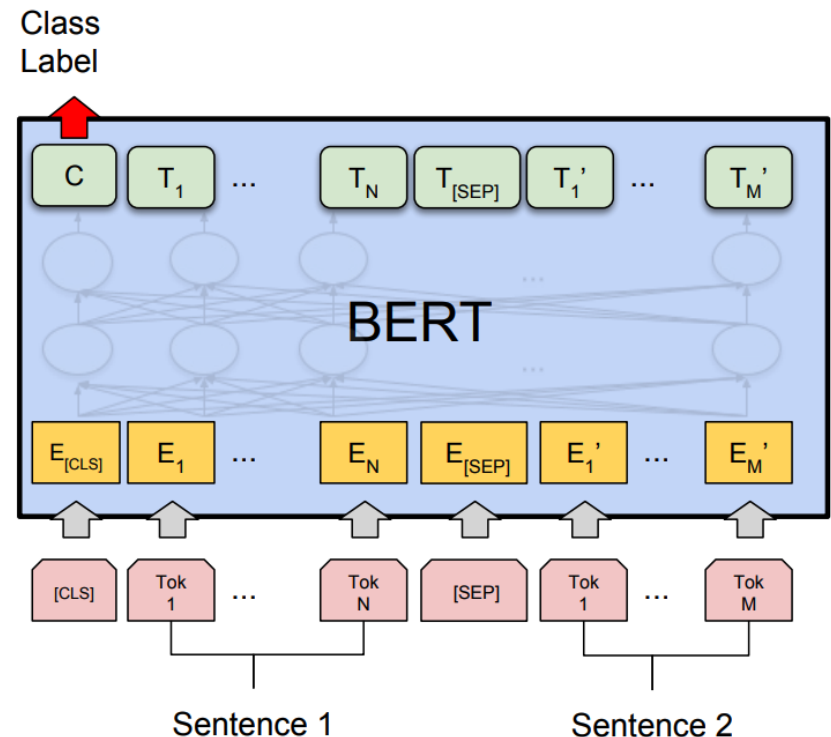
Positive:

- BERT used bi-directional (context-sensitive) embeddings.
- BERT can deal with out-of-vocabulary words.

Negative:

- BERT itself is perhaps not the best fit for pre-training NMT systems, as it does not predict words left-to-right, as most NMT systems do.
- BERT only implements the Encoder part of the Transformer model and not the Decoder
- BERT is basically a classifier
- But works for same reason m-BERT (may) work for you.

Not really!



How about more context for disambiguation in MT

- MT for Non-native speakers
- Use visual information to disambiguate MT in the metaverse
- And other topics...



About VOXReality

VOXReality's goal is to conduct research and develop new AI models to drive future XR interactive experiences, and to deliver these models to the wider European market. These new models will address human-to-human interaction in unidirectional (theatre) and bidirectional (conference) settings, as well as human-to-machine interaction by building the next generation of personal assistants.

VOXReality will develop large-scale self-supervised models that will be fine-tuned to specific downstream tasks with minimal re-training. At the same time, we will rely on modern training approaches for designing models that include subnetworks with a common

representation power but are more targeted towards specific architectures. By leveraging the once-model-for-all concept from the model training (large-scale self-supervision) and deployment (jointly learning sub-networks) perspective, we will be able to provide a catalogue of highly generic models with high representation capacity that will be efficiently specialized for downstream tasks.

Break



One day ladies will be walking their computers in the park and say, "Do you know; my little computer said a very funny thing to me this morning!"

—Alan Turing

— Alan Turing,
1951.
*Presentation at
Yale Center for
Collaborative
Arts and Media.*

Advanced Natural Language Processing

Dialogue – Chatbots – Q&A

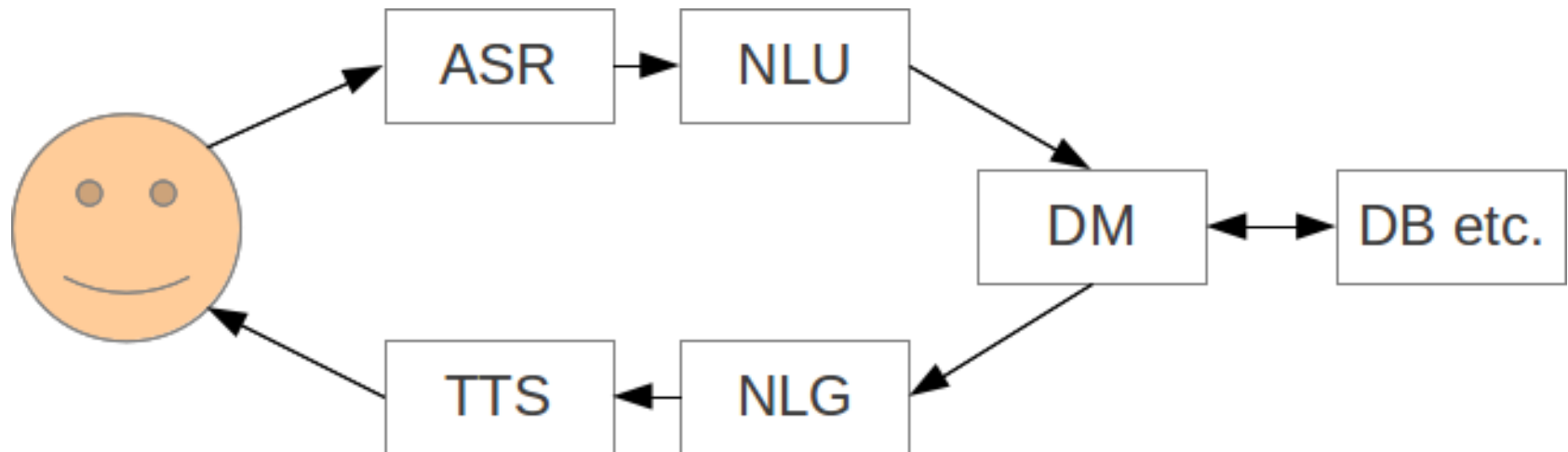
Department of Advanced Computer Sciences (DACS)
Master of AI / DSDM
Prof dr ir Jan C. Scholtes – Course year 2026-2027



Maastricht University

Dialog Modeling – the Lingo ...

Components of a (Spoken) Dialog System:

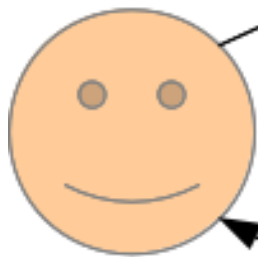


Can you identify which component might fail if a system misunderstands a user's question?

ASR = Automatic Speech Recognition
NLU = Natural Language Understanding
DM = Dialogue Management
TTS = Text-to-Speech
NLG = Natural Language Generation
DB = Data Base for background information

Dialog Modeling – the Lingo ... Today

Components of a (Spoken) Dialog System:



ChatGPT / BARD (LaMDA) / Gemini (Nano, Pro, Ultra)

ASR = Automatic Speech Recognition
NLU = Natural Language Understanding
DM = Dialogue Management
TTS = Text-to-Speech
NLG = Natural Language Generation
DB = Data Base for background information

Analyzing Human Conversations



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Human Conversations

Let's take a closer look at human conversations. We will discuss the following:

- Turns
- Interruptions
- Beginning – end-pointing
- Speech acts
- Grounding
- Conversational structure
- Sub-dialogues
 - Correction
 - Clarification
- Initiative
- Inference



A telephone conversation between a human travel agent (A) and a human client (C)

C₁: ...I need to travel in May.
A₂: And, what day in May did you want to travel?
C₃: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A₄: And you're flying into what city?
C₅: Seattle.
A₆: And what time would you like to leave Pittsburgh?
C₇: Uh hmm I don't think there's many options for non-stop.
A₈: Right. There's three non-stops today.
C₉: What are they?
A₁₀: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time.
The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the
last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.
C₁₁: OK I'll take the 5ish flight on the night before on the 11th.
A₁₂: On the 11th? OK. Departing at 5:55pm arrives Seattle at 8pm, U.S. Air
flight 115.
C₁₃: OK.
A₁₄: And you said returning on May 15th?
C₁₅: Uh, yeah, at the end of the day.
A₁₆: OK. There's #two non-stops ... #
C₁₇: #Act... actually #, what day of the week is the 15th?
A₁₈: It's a Friday.
C₁₉: Uh hmm. I would consider staying there an extra day til Sunday.
A₂₀: OK... OK. On Sunday I have ...

Properties of Human Conversation

- **Turns**

- We call each contribution a "turn" as if conversation was the kind of game where everyone takes turns.

- **Turn-taking issues**

- When to take the floor?
- When to yield the floor?

- **Interruptions and end-pointing**

- "End-pointing" is the task for a speech system of deciding whether the user has stopped talking.

What turn-taking challenges could arise in a multi-user dialog system (e.g., a chatbot in a group chat)?

URNS

C₁: ...I need to travel in May.
A₂: And, what day in May did you want to travel?
C₃: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A₄: And you're flying into what city?
C₅: Seattle.
A₆: And what time would you like to leave Pittsburgh?
C₇: Uh hmm I don't think there's many options for non-stop.
A₈: Right. There's three non-stops today.
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A₁₈: It's a Friday.
C₁₉: Uh hmm. I would consider staying there an extra day til Sunday.
A₂₀: OK... OK. On Sunday I have ...

Language as Action

Each turn in a dialogue is a kind of action

Wittgenstein (1953) and Austin (1962)



In his work *Philosophical Investigations* (1953), Ludwig Wittgenstein regularly referred to the concept of language-games. Wittgenstein rejected the idea that language is somehow separate and corresponding to reality, and he argued that concepts do not need clarity for meaning.



John L. Austin was the creator of speech act theory: He made clear that by saying something we do perform an action or just state things. He also stated that there are differences in perceiving a speech act by differentiating a speech act into locution, illocution and perlocution (<https://philpapers.org/rec/AUSHTD>).

Speech Acts (aka Dialogue Acts)

Constatives: committing the speaker to something's being the case
(*answering, claiming, confirming, denying, disagreeing, stating*)

Directives: attempts by the speaker to get the addressee to do something
(*advising, asking, forbidding, inviting, ordering, requesting*)

Commissives: committing the speaker to some future course of action
(*promising, planning, vowing, betting, opposing*)

Acknowledgments: express the speaker's attitude regarding the hearer with respect to some social action (*apologizing, greeting, thanking, accepting an acknowledgment*)

Grounding

- Participants in conversation or any joint activity need to establish **common ground**.
- Intuition: Why do elevator buttons light up?

Principle of closure. Agents performing an action require evidence, sufficient for current purposes, that they have succeeded in performing it (Clark 1996, after Norman 1988).

People need closure of their actions

Why is grounding critical for task-oriented dialogs like booking a flight?

Grounding: Establishing Common Ground

A: And you said returning on May 15th?

C: Uh, yeah, at the end of the day.

A: **OK**

C: OK I'll take the 5ish flight on the night before on the 11th.

A: **On the 11th? OK.**

C: ...I need to travel in May.

A: **And**, what day **in May** did you want to travel?

Grounding is important for computers too!

- System: Did you want to review some more of your profile?
- Caller: No.
- System: What's next?

BAD

- System: Did you want to review some more of your profile?
- Caller: No.
- System: **Okay**, what's next?

GOOD

Where can you observe
GROUNDING in modern chatbots?

Conversations have structure

- Local structure between adjacent speech acts, from the field of **conversational analysis** (Sacks et al. 1974)
- Called **adjacency pairs**:
 - QUESTION... ANSWER
 - PROPOSAL... ACCEPTANCE/REJECTION
 - COMPLIMENTS ("Nice jacket!")... DOWNPLAYER ("Oh, this old thing?")

Good for machine learning

WHY?

Another kind of structure: Sub-dialogues

Correction sub-dialogue:

Agent... OK. There's #two non-stops#

Client: #Act. . . actually#, what day of the week is the 15th?

Agent: It's a Friday.

Client: Uh hmm. I would consider staying there an extra day till Sunday.

Agent: OK...OK. On Sunday I have ...

How should a system respond if a user initiates a correction sub-dialogue while filling a slot (e.g., 'Actually, I meant...')?

Conversational Initiative

- Some conversations are controlled by one person
- A reporter interviewing a chef asks questions, and the chef responds.
- This reporter has the **conversational initiative**
- Normal human conversations, by contrast have **mixed initiative**: I lead, then you lead, then I lead.
- Mixed initiative is very hard for NLP systems, which often default to simpler styles that are much more frustrating for humans:
 - **User initiative** (user asks or commands, system responds)
 - **System initiative** (system asks user questions to fill out a form, user can't change the direction)



Even harder problems: (Knowledge) Inference

Agent: And, what day in May did you want to travel?

Client: OK uh I need to be there for a meeting that's from the 12th to the 15th.

... so, now the agent has to “inference” that the travel dates need to be before the 12th and after the 15th. Because the clients needs to be present at the meeting at the 12th until the 15th.

In GPT 4o,
how do the
latest models
deal with
complex
inference?

- How long does traveling actually takes?
- Any need to adjust for jet-leg?
- What time do the meetings start on the 12th and end on the 15th?
- ...

So, dealing with conversations is not trivial

Having a natural dialogue with a computer is challenging because it requires:

- Advanced reasoning and logic capabilities.
- Context awareness across multiple dialogue turns.
- Flexibility in handling ambiguity and implicit meanings.
- Emotional and social intelligence to interpret subtle cues.
- Grounding of abstract and philosophical concepts.

Achieving these elements requires combining cutting-edge NLP, logic, and interactive reasoning technologies. This is why we were all so impressed by ChatGPT on November 30, 2022!

Different Approaches to Dialogue Systems



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A History of Chatbots until ChatGPT

Alan Turing in the article where he proposed the question “Can machines think?” he introduced the idea for a chatbot. This article later lays out the basis for the Turing test.

In 1966 Joseph Weizenbaum, working out of the MIT AI laboratory, created ELIZA. Though ELIZA failed the Turing test, Weizenbaum laid the foundation for chatbot (then referred to as “chatterbot”) structure such as keywords, specific phrases and pre-programmed responses.

A few years later in 1972 Kenneth Colby, an American psychiatrist working out of the Stanford AI laboratory, developed PARRY, a chatbot that simulated a person with paranoid schizophrenia. PARRY did pass the Turing test with some psychiatrists.

In 1981, Jabberwacky was supposedly different from its predecessors because it used contextual pattern matching techniques and because it learnt from conversations, relying only on feedback.

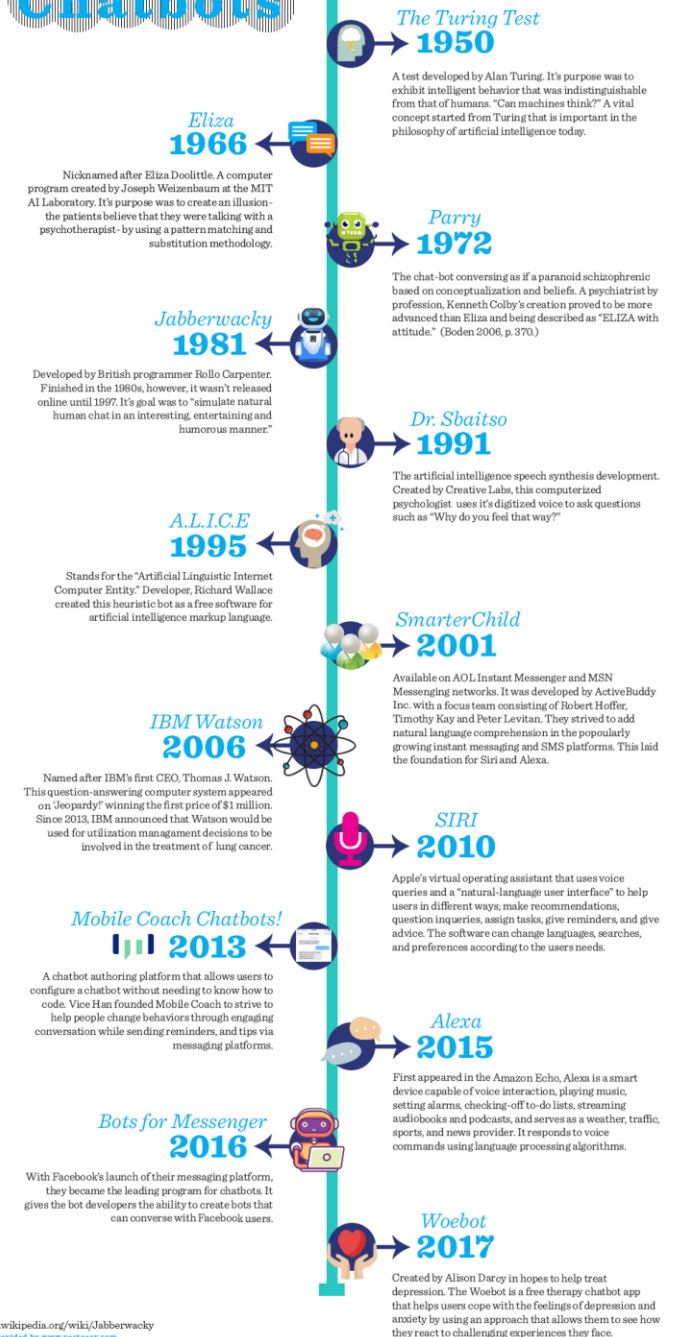
Dr. Sbaitso in 1992 was the first Chatbot using a sound card to generate speech (on MS-DOS!). Most of its responses were along the lines of “Why do you feel that way?” rather than any sort of complicated interaction.

In 1995, ALICE was another natural language processing (NLP) bot. She used sophisticated rules to match human input to carry out a conversation.

In 2001, SmarterChild was another popular chatbot used by AOL and Microsoft Messenger. It combined traditional NLP, information retrieval (on the internet), flow controls and user profiles.

Source: <https://mobilecoach.com/blog/2019/01/14/chatterbots-2-history-of-chatbots/>

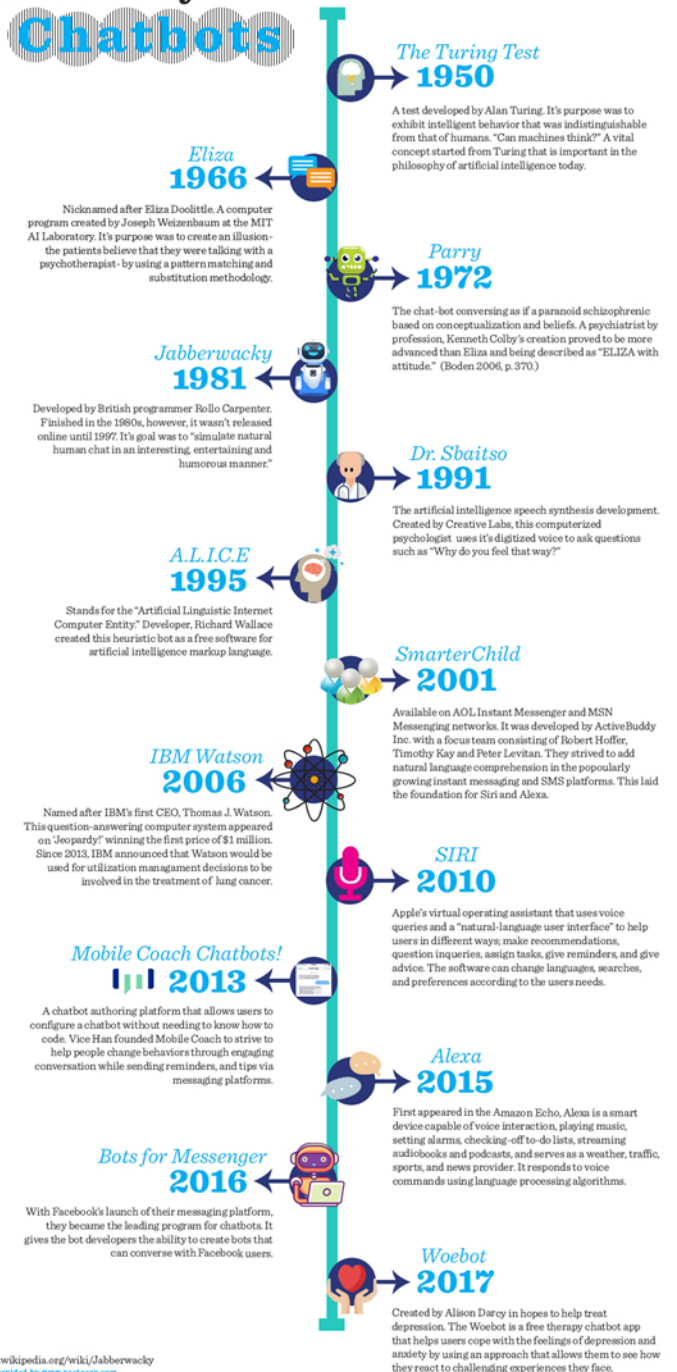
The History of Chatbots



A history of Chatbots until ChatGPT

- After, SmarterChild, IBM Watson (Ferrucci, D., Brown, E., Chu-Carroll, J., Fan, J., Gondek, D., Kalyanpur, A. A., Lally, A., Murdock, J. W., Nyberg, E., Prager, J., Schlaefter, N., & Welty, C. (2010). Building Watson: An Overview of the DeepQA Project. *AI Magazine*, 31(3), 59-79. <https://doi.org/10.1609/aimag.v31i3.2303>) combined and perfected traditional NLP, Information Retrieval, Text-Mining, Dialog flows, and speech generation.
- SIRI was originally developed with the speech recognition and generation programs from Nuance (first Lernout & Hauspie (B), then Nuance (US) and now Microsoft (US)). Apple now used Deep Neural Networks for the Speech Component of Siri. The algorithm behind the rest of Dialog algorithm is not completely clear. Although SIRI is mostly used to drive command to iOS.
- In 2013 Mobile Coach released its first chatbot, Coach Alba, that helped users achieve fitness and weight loss goals. The technology in Coach Alba was developed and expanded to become a full-featured chatbot authoring tool, which has grown into the Mobile Coach Platform.
- In the last 10 years, some of the best known chatbots have been personal assistants from tech giants like Apple (Siri – 2011), Google (Google Now – 2012), Amazon (Alexa – 2015), and Microsoft (Cortana – 2015). These voice based chatbots use NLP and sophisticated rules to perform a variety of actions and commands on multiple platforms (mobile devices, smart speakers, thermostats, etc).

The History of Chatbots



ELIZA – the mother of all Chatbots ...

The ELIZA effect, in computer science, is the tendency to unconsciously assume computer behaviors are analogous to human behaviors; that is, anthropomorphisation.

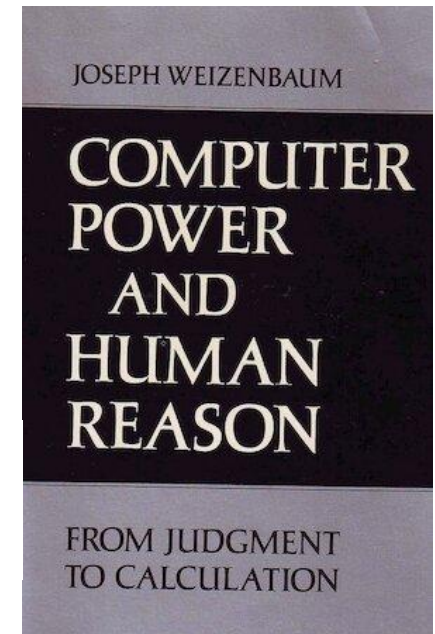
```
Welcome to

EEEEEE LL      IIII ZZZZZZ  AAAAA
EE      LL      II      ZZ  AA  AA
EEEEEE LL      II      ZZ  AAAAAA
EE      LL      II      ZZ  AA  AA
EEEEEE LLLLLL IIII ZZZZZZ  AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

See also: <https://en.wikipedia.org/wiki/ELIZA>



Eliza: Weizenbaum (1966)

Men are all alike.

IN WHAT WAY

They're always bugging us about something or other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Well, my boyfriend made me come here.

YOUR BOYFRIEND MADE YOU COME HERE

He says I'm depressed much of the time.

I AM SORRY TO HEAR YOU ARE DEPRESSED

•...

WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU?

My father

YOUR FATHER

You are like my father in some ways.

WHAT RESEMBLANCE DO YOU SEE

You are not very aggressive but I think you don't want me to notice that.

WHAT MAKES YOU THINK I AM NOT AGGRESSIVE

You don't argue with me.

WHY DO YOU THINK I DON'T ARGUE WITH YOU

You are afraid of me.

DOES IT PLEASE YOU TO BELIEVE I'M AFRAID OF YOU

My father is afraid of everybody.

WHAT ELSE COMES TO MIND WHEN YOU THINK OF YOUR FATHER

Bullies.

DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT YOUR
BOYFRIEND MADE YOU COME HERE

How does it work?

- Eliza uses this one weird trick

That trick: be a Rogerian psychologist

- Draw the patient out by reflecting patient's statements back at them
- Rare type of conversation in which one can “assume the pose of knowing almost nothing of the real world”

Rogerian psychologist

Patient: "I went for a long boat ride"

Psychiatrist: "Tell me about boats"

- You don't assume she didn't know what a boat is
- You assume she had some conversational goal
- Chatbots trying to pass the Turing test usually choose some such domain

Eliza pattern/transform rules

(0 YOU 0 ME) [*pattern*]

(WHAT MAKES YOU THINK I 3 YOU)
[*transform*]

0 means *

The 3 is the constituent # in pattern

You hate me

WHAT MAKES YOU THINK I HATE YOU



Eliza Rules

function ELIZA GENERATOR(*user sentence*) **returns** *response*

Find the word w in *sentence* that has the highest keyword rank

if w exists

Choose the highest ranked rule r for w that matches *sentence*

$response \leftarrow$ Apply the transform in r to *sentence*

if $w = \text{'my'}$

$future \leftarrow$ Apply a transformation from the 'memory' rule list to *sentence*

Push *future* onto memory stack

else (no keyword applies)

either

$response \leftarrow$ Apply the transform for the NONE keyword to *sentence*

or

$response \leftarrow$ Pop the top response from the memory stack

return(*response*)

NONE

- PLEASE GO ON
 - THAT'S VERY INTERESTING
 - I SEE
-
- ... TELL ME ABOUT YOUR MOTHER ...

Memory

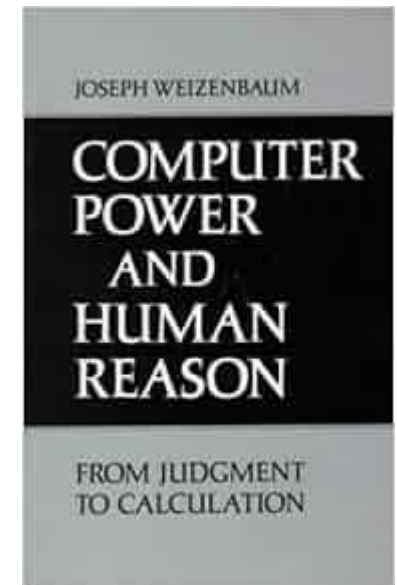
- (MEMORY MY
(0 YOUR 0 = LETS DISCUSS FURTHER WHY YOUR 3)
- (0 YOUR 0 = EARLIER YOU SAID YOUR 3)
- Whenever “MY” is highest keyword
 - Randomly select a transform on the MEMORY list
 - Apply to sentence
 - Store on a stack
- Later, if no keyword matches a sentence
 - Return the top of the MEMORY queue instead
- (Earliest proposal for a hierarchical model of discourse!)

Try out ELIZA here: <https://web.njit.edu/~ronkowitz/eliza.html> or in the tutorial.

Ethical implications: Anthropomorphism and Privacy

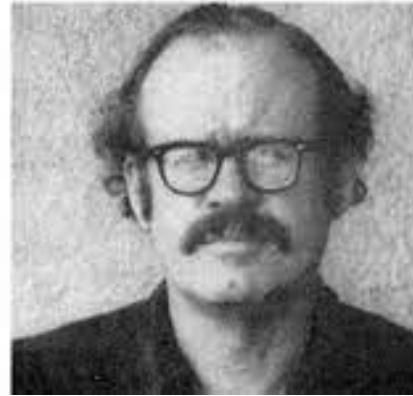
- People became deeply emotionally involved with the program.
- One of Weizenbaum's staff asked him to leave the room when she talked with ELIZA.
- When he suggested that he might want to store all the ELIZA conversations for later analysis, people immediately pointed out the privacy implications, suggesting that they were having quite private conversations with ELIZA

In reaction to how people responded to Eliza, he wrote this book. Still relevant today!

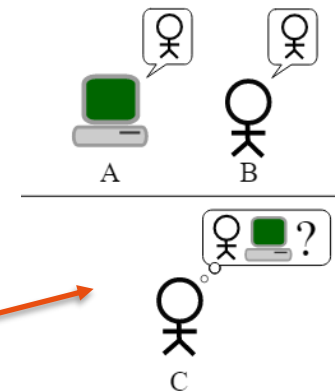


Parry: the first system to pass the Turing Test

- PARRY was written in the early 1970s by psychiatrist Kenneth Colby, then at Stanford University. While ELIZA simulated a Rogerian therapist, PARRY attempted to simulate a paranoid schizophrenia.
- The program implemented a crude model based on concepts, conceptualizations, and beliefs.
- It embodied a conversational strategy, and as such was a much more serious and advanced program than ELIZA. It was described as "ELIZA with attitude".
- PARRY was tested using a variation of the Turing Test. A group of experienced psychiatrists analyzed a combination of real patients and computers running PARRY through teleprinters. Another group of 33 psychiatrists were shown transcripts of the conversations. The two groups were then asked to identify which of the "patients" were human and which were computer programs. The psychiatrists were able to make the correct identification only 48 percent of the time — a figure consistent with random guessing.



H: Good afternoon. My name is Doctor Jones.
P: Pleased to meet you.
H: How are you?
P: All right I guess.
H: Could you tell me your name?
P: Pat Smith.
H: I thought you might have some problems.
P: I've been upset lately.
H: By what?
P: People get on my nerves sometimes.



The first system to pass the Turing test (in 1971)

Parry's persona

- 28-year-old single man, post office clerk
- No siblings and lives alone
- Sensitive about his physical appearance, his family, his religion, his education and the topic of sex.
- Hobbies are movies and gambling on horseracing,
- Recently attacked a bookie, claiming the bookie did not pay off in a bet.
- Afterwards worried about possible underworld retaliation
- Eager to tell his story to non-threatening listeners.



The Atlantic

When PARRY Met ELIZA: A Ridiculous Chatbot Conversation From 1972

In January 1972, as a demonstration during an international computer conference, the computer science pioneer Vint Cerf decided to take the bots to their logical conclusion: Using ARPANET, he set up a conversation between ELIZA and PARRY. It was a bicoastal meeting of the computer minds: ELIZA was based at MIT, PARRY at Stanford.

ELIZA: Good Evening. Tell me your problems. Please terminate input with a period or a question mark.

PARRY: People get on my nerves sometimes.

ELIZA: I am not sure I understand you fully.

PARRY: You should pay more attention.

ELIZA: Suppose you should pay more attention.

PARRY: You're entitled to your own opinion.

ELIZA: What makes you think I am entitled to my own opinion?

PARRY: It bothers me just to be around people in general.

ELIZA: Please go on.

PARRY: I went to the track at Bay Meadows a while back.

ELIZA: What does that suggest to you?

PARRY: I went to the races.

.....

ELIZA: Would you like to discuss your nerves?

PARRY: I guess so.

ELIZA: What does that suggest to you?

PARRY: What are you getting at?

ELIZA: What answer would please you most?

PARRY: It's not a matter of like or dislike.

ELIZA: What else would you like to discuss?

PARRY: I am getting tired of answering the same questions over and over.

ELIZA: What resemblance do you see?

PARRY: Why do you need to know that?

ELIZA: What do you think?

PARRY: I don't understand your motives.

ELIZA: Why don't you understand my motives?

And Then There was ChatGPT ...



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ChatGPT Major Contributions

1. Use language like a human being.
2. Stay* within human acceptable boundaries.
 - LLMs can generate outputs that are untruthful, toxic, or simply not helpful to the user.
 - ChatGPT is aligned with user intent on a wide range of tasks by fine-tuning with human feedback.
 - Still simple mistakes, the results show that fine-tuning with human feedback is a promising direction for aligning language models with human intent.

*) When used as intended...

Training language models to follow instructions with human feedback. L Ouyang, J Wu, X Jiang, D Almeida, C Wainwright... - Advances in Neural Information Processing Systems, March 2022.
<https://arxiv.org/abs/2203.02155>

How robots can learn to follow a moral code. Ethical artificial intelligence aims to impart human values on machine-learning systems. Neil Savage. Nature Outlook. October 26, 2023.
<https://www.nature.com/articles/d41586-023-03258-1>

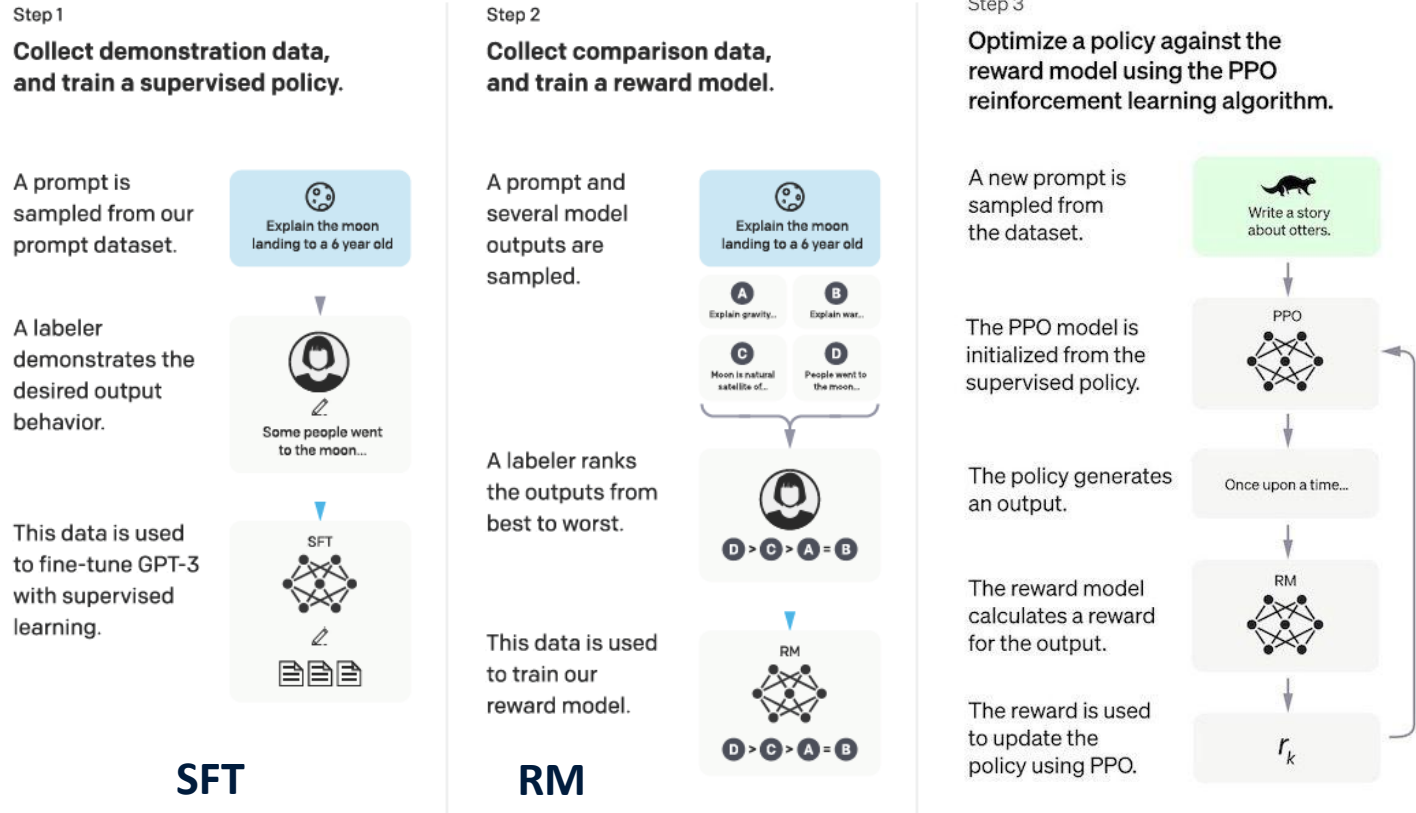
Aligning ChatGPT with Human Values and Norms

- 
- Starting with a set of labeler-written prompts and prompts submitted through a language model API, a dataset is collected of labeler demonstrations of the desired model behavior, which we use to fine-tune GPT-3 using supervised learning.
 - We then collect a dataset of rankings of model outputs, which we use to further fine-tune this supervised model using reinforcement learning from human feedback. We call the resulting models InstructGPT.
 - In human evaluations on our prompt distribution, outputs from the 1.3 B parameter InstructGPT model are preferred to outputs from the 175B GPT-3, despite having 100x fewer parameters.
 - Moreover, InstructGPT models show improvements in truthfulness and reductions in toxic output generation while having minimal performance regressions on public NLP datasets*.

*) We will get back this later. Numerically, the results are similar, however, the errors made can be harmful as the models leave out essential facts and make up non-sense.

Aligning LLMs with Human Values

PPO



See the
Lecture on
Fine Tuning
LLM for more
details on
this process

Figure 2: A diagram illustrating the three steps of our method: (1) supervised fine-tuning (SFT), (2) reward model (RM) training, and (3) reinforcement learning via proximal policy optimization (PPO) on this reward model. Blue arrows indicate that this data is used to train one of our models. In Step 2, boxes A-D are samples from our models that get ranked by labelers. See Section 3 for more details on our method.

Testing LLM ChatBots

THE WALL STREET JOURNAL. /le Real Est

Can AI Replace Humans? We Went to the Fast-Food Drive-Through to Find Out

Chatbots are already working in restaurants across the U.S. Our columnist put in about 30 orders at a Hardee's.



Across the country AI chatbots are now taking fast-food drive-through orders. WSJ's Joanna Stern put the tech through a series of tests at a Hardee's—including blasting dog barking sounds and asking some crazy questions.

- See: https://www.wsj.com/articles/can-ai-replace-humans-we-went-to-the-fast-food-drive-through-to-find-out-193c03e9?mod=hp_featst_pos4

Other Considerations

Unanswerable Questions

Why LLMs Often Add a Summary at the End of a Prompt Responds

- **LLMs like GPT-4 operate within a limited context window** (e.g., 8K, 32K, or 128K tokens). To retain information over long interactions, the model can create a summary of the previous answer into the next prompt. This enables the model to preserve the essence of earlier exchanges without consuming too much of the available context space. This is particularly important in multi-turn conversations, where the model cannot recall earlier outputs once the context window fills up.
- **Self-reflection and output validation:** A summary at the end of a response helps the model to summarize its own reasoning (e.g., in Chain-of-Thought prompting) and to produce a “verifiable conclusion”. In addition, it allows the model to check for consistency between the question and the generated answer
- **Input for external tools or agents:** in agent-based systems, the model's output is often forwarded to another tool or module or it can be stored as an internal “thought” or memory or used as input for planning and task decomposition. In such cases a short summary is better for the calculations of the subsequent agents or subtasks.
- **Easier logging or display to users:** in chat interfaces (such as ChatGPT), a short summary (e.g., a conversational turn title) is often generated to help users maintain an overview—especially in long conversations or when browsing past interactions.



There are also Unanswerable Questions

- Challenges:
 - System invent answers if they cannot answer
- Data Augmentation on Part of Squad 2.0 data set

Article: Endangered Species Act

Paragraph: “ ... Other legislation followed, including the Migratory Bird Conservation Act of 1929, a 1937 treaty prohibiting the hunting of right and gray whales, and the Bald Eagle Protection Act of 1940. These later laws had a low cost to society—the species were relatively rare—and little opposition was raised.”

Question 1: “Which laws faced significant opposition?”

Plausible Answer: later laws

Question 2: “What was the name of the 1937 treaty?”

Plausible Answer: Bald Eagle Protection Act

Figure 1: Two unanswerable questions written by crowdworkers, along with plausible (but incorrect) answers. Relevant keywords are shown in blue.

Questions?



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Exercise 1: NLP Summarization Techniques (Extractive vs. Abstractive)

Summarization is a core downstream task for ANLP models, divided historically and architecturally into extractive and abstractive methods.

- Define **Extractive Summarization** and provide two classical (pre-Transformer) algorithms used to score and select sentences from an input document (e.g., as mentioned on slides 4 and 5).
- Define **Abstractive Summarization**. Why do Sequence-to-Sequence (Encoder-Decoder) architectures excel at this task compared to early Generative Grammars?
- According to the medical evidence summarization study by Tang et al. (2023) shown in this lecture, what are two major risks or failure modes (e.g., hallucinations, over-confidence) associated with using LLMs like ChatGPT for abstractive summarization of long, highly technical texts?

Exercise 2: Transformer Architectures for Sequence-to-Sequence Tasks (T5, PEGASUS, BART)

While BERT (Encoder-only) and GPT (Decoder-only) dominate specific niches, Encoder-Decoder architectures are natively suited for transduction tasks like abstracting and translation.

- **T5 (Text-to-Text Transfer Transformer)** formulates all NLP tasks into a unified text-to-text format. Give two examples of how input prompts are prepended with specific prefixes to switch T5's behavior between translation and summarization.
- The **PEGASUS** model uses a specific self-supervised pre-training objective tailored for summarization. Describe the "Gap Sentence Generation" (GSG) objective. How does masking entire important sentences (rather than random individual tokens) better prepare the model for abstractive summarization?
- **BART** combines a bidirectional encoder with an autoregressive decoder. How is BART pre-trained as a "denoising autoencoder" using text infilling, and how does this differ from standard BERT masking?

Exercise 3: Human-Computer Dialogue and the ELIZA Effect

The history of chatbots reveals the persistent psychological tendency of humans to anthropomorphize machines, even when interacting with simple pattern-matching scripts.

- Describe the core mechanism behind Joseph Weizenbaum's **ELIZA** (1966). How did ELIZA use "pattern/transform rules" (e.g., mapping "O YOU O ME" to "WHAT MAKES YOU THINK I 3 YOU") and a memory queue to simulate a Rogerian psychotherapist?
- Define the "**ELIZA Effect**". Based on the lecture, provide one historical example of how users reacted to ELIZA (e.g., regarding privacy or emotional attachment) that alarmed Weizenbaum.
- Compare ELIZA to **PARRY** (1972). While ELIZA acted as a passive listener reflecting statements back at the user, what complex persona and internal states (e.g., beliefs, fears of underworld retaliation) did PARRY simulate to pass a modified Turing Test administered to psychiatrists?

Discussion



On Machine Translation: When Translation Is Free, Who Answers for What It Says?

Rule-based translation never reached acceptable quality; statistical translation did, within a domain; neural translation is now good enough that for most texts nobody pays a translator. Translation has become free and instant.

A human translator does more than map sentences: they decide what the author meant, what the reader needs, what cannot be said in the target language and what to do about it. They can be asked, and they can be blamed. Where did those decisions go?

When a machine-translated contract, verdict or medical instruction turns out to say something different in the other language, who is responsible? And if nobody is, has something been lost that no BLEU score would ever show?

On the ELIZA Effect: Whose Illusion Is It, and Who Is Responsible for It?

In 1966 a few hundred lines of pattern-matching code made people confide in a machine, ask for privacy, and feel understood. Weizenbaum was horrified, not by the program but by the people. Sixty years later, the program is a trillion parameters and the people have not changed.

Whose illusion is the ELIZA effect: a defect in the machine, which only simulates, or a feature of humans, who are built to find a mind behind language? If the second, is it a weakness to be corrected, or the same capacity that lets us understand each other at all?

Modern chatbots are optimised for exactly the responses that trigger the effect. Is it wrong to build systems that exploit a human reflex, and if the system is helpful, who is harmed?

On Summaries: Is There Such a Thing as a Faithful Summary?

An extractive summary copies sentences and cannot say anything the source did not; an abstractive summary writes new sentences and sometimes says things the source never did. For medical evidence the second is dangerous. But the first is not innocent either: choosing which sentences to keep is already a claim about what matters.

Is there such a thing as a faithful summary? Every summary is shorter than its source, so every summary leaves things out, and leaving out is a judgement. Is fidelity a property of the text, or of the relation between a summary and the purpose of a particular reader?

If a summary is always an argument about what matters, then a summariser, human or machine, is making that argument on someone's behalf. Who checks it, and is "it reads well" evidence of anything?

